

ESP Study - Conclusions

Malec

ESP study conclusions

Scenario 1: A person in my group got 15 out of 50 Zener cards correct, a proportion of 0.30. If we make the skeptic's assumption that they were just guessing, does this provide a plausible explanation for their result of 15 correct cards?

The dotplot depicts the results of 1000 simulations of the a 50-card Zener card test under the assumption that the person selecting the cards is simply guessing. Each dot in the dotplot represents the proportion of cards correctly selected in one 50-card test. The distribution is symmetric and bell-shaped. The proportion of correct guesses is about .2, give or take 0.06. Most simulations resulted in proportions between about 0.1 and 0.3, and almost all resulted in between about 0.08 and 0.32 of the cards guessed correctly.

Under the assumption that my classmate was just guessing, getting 15 or more (or 5 or fewer) cards correct happens about 13% of the time ($p\text{-value} = 0.1260$). This means getting a score as or more extreme than 15 is not that unusual. This result is compatible with the null model - that is, it is plausible that my classmate was simply guessing (and has no psychic ability).

Count samples

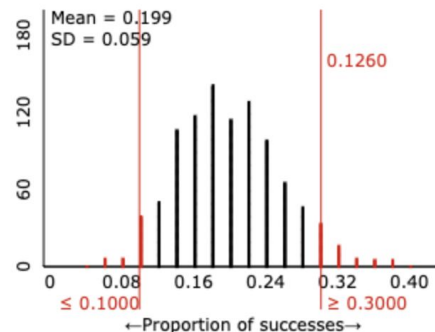
As extreme as

Proportion of samples:
(55 + 71) / 1000 = 0.1260

Options:

- ☒ Two-sided
- (between: ☐)
- ☐ Exact Binomial
- ☐ Normal Approximation

☒ Summary Statistics

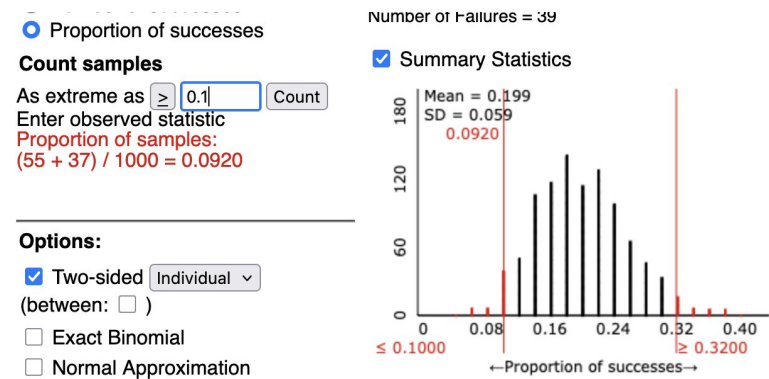


ESP study conclusions

Scenario 2: A person in my group got 5 out of 50 Zener cards correct, a proportion of 0.1. If we make the skeptic's assumption that they were just guessing, does this provide a plausible explanation for their result of 5 correct cards?

On average, a person guessing will get about 10 out of 50 cards correct. Under the assumption that my classmate was just guessing, getting a result as or more extreme than 5 cards correct means either less than or equal to $5/50=0.1$ OR greater than or equal to about $15/50=0.3$. (NOTE: the sharp-eyed among you will notice the graph to the right puts the right-tailed red line at 0.32, which is not a mistake - I'll mention this Wednesday)

This happens about 10% of the time ($p\text{-value} = 0.0920$). This means that if my friend was simply guessing, getting a score as or more extreme than 5 is uncommon, but not all that unusual. We cannot say for sure that my friend was not guessing. (i.e. here, we cannot comfortably reject the null hypothesis.)



ESP study conclusions

Scenario 3: A person in my group got 19 out of 50 Zener cards correct, a proportion of 0.38. If we make the skeptic's assumption that they were just guessing, does this provide a plausible explanation for their result of 2 correct cards?

Under the assumption that my classmate was just guessing, getting 19 or more (or 2 or fewer) cards correct happens only about 0.6% of the time ($p\text{-value} = 0.0055$). This means getting a score as or more extreme than 19/50 is extremely unusual if my friend was simply guessing. We have evidence to suggest that my friend was not simply guessing, and we should investigate further the theory that my friend has psychic abilities. (NOTE: we CANNOT conclude that my friend is psychic. All we can say is that their score on the Zener test is not compatible with the idea that they were just guessing.)

Second note: it probably doesn't seem obvious that 19 or more is equivalent to 2 or fewer, but I'll give you a bit of justification for that next class. You can trust the computer.

☒ Proportion of successes

Count samples

As extreme as $\geq$ 0.3800

Proportion of samples:
 $(2 + 9) / 2000 = 0.0055$

Options:

☒ Two-sided

(between: ☐)

☐ Exact Binomial

☐ Normal Approximation

PROPORTION OF SUCCESSSES = 0.38

☒ Summary Statistics

